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Lightning response to temperature and aerosols

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Abstract

Lightning has profound social implications for public safety and usually causes casualties and significant damage to infrastructure. Due to the advancement of both ground-based and spaceborne detection technology, lightning has been monitored globally in recent decades as an indicator of severe weather and an essential variable of climate change. This article reviews recent progress in the study of lightning variations and their response to temperature and aerosols, based on both spaceborne and ground-based lightning data. The responses of lightning to temperature and aerosol show large spatial and temporal heterogeneity, with relation to the meteorological and environmental conditions. The latest research show that lightning exhibited significant increase in some high altitude or high latitude regions, such as the Tibetan Plateau and Arctic regions, where undergone fast warming during the recent decades and the ecosystems is fragile. Aerosol particles play an important role in modulating lightning variations under certain dynamical and thermodynamic conditions in some regions, even on a global scale. The projected lightning activity will generally increase in the future but may with very few exceptions. Continuous long-term lightning observations with consistent spatial and temporal detection efficiency remain crucial for tracking the response of lightning to climate change in the coming decades.

1. Introduction

Lightning is a natural hazard and not only leads to significant loss of life and property (Yair 2018; Holle 2016; Qie et al 2023), but also triggers wildfires producing large scale emissions of carbon dioxide, methane and other gases including non-methane volatile organic compounds, thereby affecting the evolution of species and ecosystems of the earth system (Abatzoglou et al 2016; Veraverbeke et al 2017; Hessilt et al 2022). Lightning is also a large driver of climate change by producing greenhouse gases, and it is the only natural source of nitrogen oxides (NO_x) in the troposphere, modulating the oxidation capacity of the upper troposphere and regulating tropospheric chemistry (Price et al 1997; Brune et al 2021; Mao et al 2021; Liu et al 2024). A strong correlation has also been found between global lightning activity and upper

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tropospheric water vapor and chemistry (Price, 2000; Price et al 2023; Xu et al, 2023),
key elements of Earth's climate.

Lightning is generated within thunderstorms, which are deep convective weather
systems developed in unstable humid and hot atmospheres. As a result, lightning has
usually been regarded as an indicator of severe weather phenomena such as flash floods,
extreme hail and tornadoes. Recently, lightning has been declared to be an essential
climate variable, based on a recommendation by the Task Team on Lightning Observa-
tion for Climate Applications as part of the Atmospheric Observation Panel for Climate
of the WMO and the Global Climate Observing System (Aich et al 2018; Füllekrug et
al 2022). All these motivate efforts to investigate the spatial and long-term variations
of lightning in the context of global climate change. In this article, we will briefly
review the recent research progress in lightning variations and its responses to
temperature and aerosols.

2. Global lightning distribution and regional differences

The importance of lightning in the Earth system is becoming evident. In recent
decades, ground- and space-based lightning detection and location technologies have
largely extended the understanding of lightning distribution in global and regional
scales, as well as in many time scales. To evaluate how global lightning responds to
temperature and aerosol changes, it is necessary to understand the general behavior and
global distribution of lightning in the current global climate.

2.1 Global lightning climatology and hotspots

Lightning activity exhibits significant spatial and seasonal heterogeneity globally.
Based on the Optical Transient Detector (OTD) data, Christian et al (2003) presented
the first satellite-based lightning climatology and revealed the geographical distribution
and seasonal variations of lightning activity on a global scale. They found a dominant
Northern Hemisphere summer peak in the annual cycle of lightning flash density and
the evidence of tropically driven semiannual lightning cycles, largely controlled by
temperature and water vapor. The global annual average flash rate was 44 ± 5 flashes

s^{-1} . Cecil et al (2014) constructed the gridded lightning climatology based on TRMM-LIS (1998-2010) and OTD (1995-2000) data (Fig. 1). The mean global flash rate from the merged datasets varies from around 35 flashes s^{-1} in February (austral summer) to 60 flashes s^{-1} in August (boreal summer) with a mean rate of 46 flashes s^{-1} .

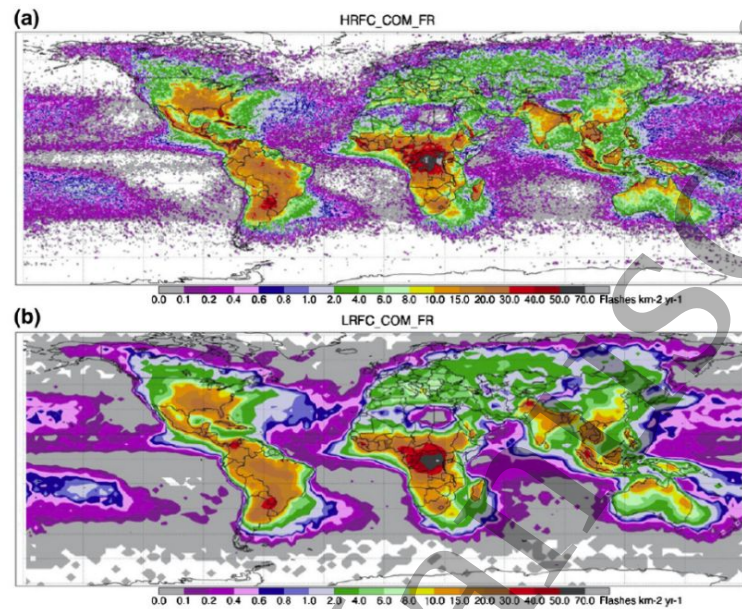


Fig. 1 Mean annual flash rate from merged LIS and OTD data. (a) at 0.5° grid, and (b) at 2.5° grid (Adopted from Cecil et al 2014).

Albrecht et al (2016) identified the Earth's lightning hotspots and other regional features based on a very high-resolution total lightning climatology gridded at 0.1° from LIS. Surprisingly, they found that the earth's top-ranked total lightning hotspot locates over Lake Maracaibo in Venezuela (9.75° , -71.65°) with a flash rate density of 233 flashes $km^{-2} yr^{-1}$, and the previously believed hotspot of Congo basin in Africa (Christian et al 2003; Cecil et al 2014) descend to second place with 205 flashes $km^{-2} yr^{-1}$. In Asia, the southern slopes of the Himalayas can be regarded as another lightning hotspot region with maximum flash rate density of 143.1 flashes $km^{-2} yr^{-1}$ in Daggar, Pakistan (34.45° , 72.35°), the westernmost concave indentation region of Himalayas foothills in Pakistan because of the Himalayas topographic lifting to the air mass flows from Arabian Sea and Bay of Bengal (Houze et al 2007; Romatschke et al 2010; Qie et al 2014; Wu et al 2016). They also found that Africa is represented by 283 of the top 500 places with the highest flash rate, followed by Asia with 87, South America with 67, North America with 53, and Oceania with 10.

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Peterson et al (2021), taking the horizontal scale of lightning into account, found the distribution of flash extent density remains unchanged from the flash rate density using the data from LIS/OTD and LIS on the International Space Station (ISS-LIS). Many of the hotspots in flash extent density including the top hotspot of Lake Maracaibo shows the same pattern with the flash rate density. It should be noted that Zipser et al (2006) indicated the major areas of the strongest thunderstorms on earth to be located in the south-central US, south-eastern South America (Argentina) and equatorial Africa (Congo basin), with season-dependent maxima also found in northern Australia, the Indian subcontinent and the Sahel. Recently, Peterson (2024) reported that maritime thunderstorms exhibit the highest lightning densities, in coastal regions in the Gulf of Mexico and east of South-Africa. Such oceanic storms possess all the meteorological attributes of their land-based mesoscale convective systems, but on a smaller horizontal scale, leading to concentrated electrical activity with high flash rates.

Holzworth et al (2019) found that the global lightning superbolt distribution, defined by the World Wide Lightning Location Network (WWLLN) as having stroke energy above 1 MJ in the very low frequency band, peaks in the Northern Hemisphere winter, and is quite different from the distribution of total lightning mentioned above. The superbolts are more common over oceans than land in the wintertime, a period when the ocean is warmer than land, further lending support to the temperature dependence of lightning. The WWLLN Global Lightning Climatology shows important variability of lightning strokes on interannual timescales (Kaplan and Lau, 2021). The lightning climatology from the WWLLN appears to be consistent with in situ rainfall observations and with the TRMM rainfall climatology (Virts et al 2013).

2.2 Land-ocean contrast in lightning activity

There are clear and sharp regional differences in flash density and flash rate in response to surface temperature and water vapor availability. The spatial distribution of annual lightning distribution is largely dominated by continental areas in the tropics with three clear chimney regions, Africa, the Maritime Continent (SE Asia), and the

South American. The average land/ocean lightning ratio is approximately 10:1 (Christian et al 2003; Cecil et al 2014). From a thermodynamical perspective, the land is warmer, and its stability conditions are more conducive to the development of thunderstorms. However, the corresponding land–ocean contrast in the values of the convective available potential energy (CAPE), which is widely used to evaluate conditional instability of atmosphere, is only qualitatively coincident with the observed lightning distribution. Upon closer examination, the CAPE contrast is insufficient to explain the order-of-magnitude contrast in lightning prevalence between the continents and the oceans (Williams and Stanfill, 2002; Romps et al 2018).

Although thermodynamics may play the leading role in the land/ocean lightning/thunderstorms contrast, the aerosol contribution could be also important. From a microphysical perspective, aerosol particles providing necessary cloud condensation nuclei (CCN) for cloud droplets (Rosenfeld et al 2008; Fan et al 2018) also shows a marked land-ocean contrast, with more severe pollution over land. Moreover, Pan et al (2022) suggested that giant sea salt nuclei suppress lightning in oceanic convection. This difference is manifested in charge separation processes and eventually in the lightning activity of thunderstorms developing in clean or polluted conditions, as discussed in section 4 below.

It is expected that both thermodynamics and aerosol effects play significant role in this prominent land/ocean lightning contrast because their impact on updrafts and microphysical processes of the thunderstorm, but their respective quantitative contributions still need to be addressed. Other factors may also contribute to the land-ocean contrast, including the terrain topography and the cloud base height, and as well as the different shapes of updraft buoyancy profiles over land compared to the ocean (Williams and Stanfill, 2002; Romps et al 2018).

2.3 Regional lightning distributions

The LIS/OTD and WWLLN datasets have made it possible to study lightning distributions in many specific regions, such as the Tibetan Plateau with an average

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elevation higher than 3500 m (Qie et al 2003, 2022), South Asia and the Indian subcontinent (e.g., Kumar and Kamra 2012a; Qie K., et al 2021), Congo (Soula et al 2016), and the Northwest Pacific region (Zhang et al 2018), to name a few. The cloud-to-ground (CG) lightning distribution has also been studied in many regions, the United States (e.g., Orville and Huffines 2001; Holle et al 2016), China (e.g., Xu et al 2022; Zheng et al 2016), Brazil (e.g., Pinto et al 2009), Europe (e.g., Enno et al 2020), the eastern Mediterranean (Ben-Ami et al 2015) and many other regions based on CG lightning location networks. All these results also showed obvious diurnal and seasonal variations in lightning activity.

Regional lightning activity exhibits a clear response to the environmental thermodynamics. For example, Qie et al (2003a, b) found an obvious thermodynamic control on lightning, and lightning rate nonlinearly response to the CAPE. The sensible heat flux plays important role, at least in the Tibetan Plateau, in modifying the efficiency of generating lightning from cloud buoyancy (Toumi and Qie, 2004). Fuchs et al (2015) indicated the environmental controls on storm intensity and charge structure in multiple regions of the continental United States.

Although lightning activity occurs primarily in tropical and subtropical land areas, lightning has indeed been detected in colder polar regions. The NOAA GSOD (Global Summary of Day) dataset on thunder day observations in recent decades indicates that 390 ground stations north of the Arctic Circle have showed at least one lightning day in their records (Williams et al 2022).

3. Lightning variations in a warming climate

Studies have shown that lightning has a sensitive response to surface temperature, and that global lightning can be regarded as an indicator of climate change (Williams, 1992, 2005; Reeve and Toumi, 1999; Price 2013). However, when checking lightning dependence on temperatures on a global scale, the increase in lightning activity may not be necessarily so profound in the context of climate warming, due to the compensating effect of different regions exhibiting opposite trend. Most of the progress

in quantifying the relationship and sensitivity of lightning activity to the atmospheric temperature have been made on a regional scale.

3.1 Lightning trends based on observational data

In the context of global climate change, there is an increasing interest in understanding the global response of lightning to temperature changes at different time scales. Williams et al (2022) discussed in detail the lightning variations with temperature on the time scales of the diurnal, semiannual, annual, decadal, multi-decadal, and ENSO timescales. Here we briefly survey the key findings on trends in lightning response to changes in temperature.

Based on LIS/OTD lightning data from 1996 to 2013, Qie K et al (2020) identified significant trends in lightning density changes in some specific regions over the TRMM domain, with Southeast North America, central South America and eastern Australia significantly decreasing, and northwest South Asia increasing, while no significant trend was found in the global scale. The control factors of lightning variation trends vary in different regions. The negative lightning trends in three regions are found to be more likely caused by decreases of the CAPE, while the positive trend seems to be related to increase of aerosol in northwest South Asia.

Koshak et al (2015) found a clear conspicuous decrease in CG lightning over the continental US in the period 2003–2012 with a 12% decline from 2003–2007 compared to 2008–2012 values. The wet bulb temperature over the Continental US for the same period had also decreased while the dry bulb temperature increased. However, the total lightning activity from the LIS did not exhibit a significant trend over the same period.

The Arctic region is one of the most sensitive areas to climate change. Holzworth et al (2021) found the Arctic was becoming more influenced by lightning and showed that the fraction of WWLLN strokes above 65°N relative to the total global number of lightning strokes had increased by a factor of 3 as the temperature anomaly increased from 0.65°C to 0.95°C from 2010 to 2020. The Arctic is warming faster than the global average, about four times greater than the rest of the Earth, which caused the rapid disappearance of Arctic Sea ice (Pithan and Mauritsen, 2014). Lightning changes may

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affect the very fragile ecosystem in the Arctic region. Saha et al. (2023) suggested that it is the loss of summer sea-ice in the Arctic that drives lightning activity, however Popykina et al. (2024) showed that the observed increase in lightning in northern polar latitudes can be attributed to a few convective storms each summer, and is dominated by the contribution of north Siberia, where land heating is most pronounced. This trend is likely to increase with additional warming of Eurasia.

Likewise, the Tibetan Plateau is also exhibiting significant warming in recent decades (Yao et al 2019) and the most significant temperature rise was found to occur in the 1990s (Gao et al 2015) with a rate of $0.25^{\circ}\text{C decade}^{-1}$ from 1998 to 2013 (Duan and Xiao, 2015). In such a context, Qie et al (2022) identified a significant increase in lightning activity in the eastern part of the Tibetan Plateau by using LIS/OTD (1997-2013) and WLLN (2005-2019) lightning data. Interestingly, the thunderstorm frequency in that area during the same period of time showed an increase while the thunderstorm intensity did not, suggesting that the increase in thunderstorm number caused the increasing in lightning activity. These trends are supposed to be associated with the special thermodynamic environment over the Tibetan Plateau. The unstable energy is easily released to trigger ordinary thunderstorms, but it is difficult to accumulate to generate severe thunderstorms.

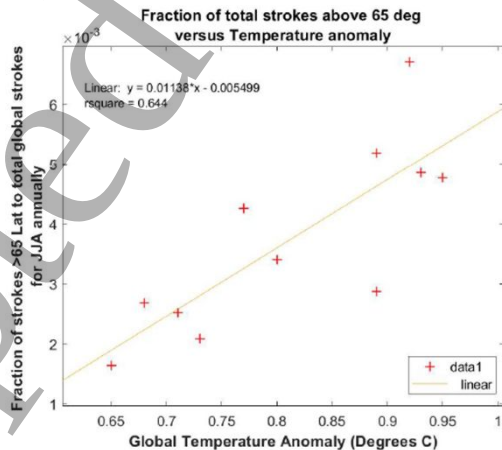


Fig. 2 Lightning flash density increase in a warming climate. Fraction of WLLN strokes above 65°N and total WLLN strokes and temperature anomaly (Adopted from Holzworth et al 2021).

The El-Nino Southern Oscillation (ENSO) is an irregularly repeated interannual variation alternating between warm (El Niño) and cold (La Niña) phases, based on sea surface temperature (SST) measurements. Both types of ENSO episodes have a fundamental impact on global climate, with a warm El Niño period may lead to more drastic changes (Allen et al 2000). The effect of ENSO on lightning activity was found in the Maritime Continent (including southeast Asia, Indonesia, and tropical Australia) with a clear prevalent tendency in the warm El Nino phase (e.g., Hamid, et al 2001; Sători et al 2009; Yuan et al 2016). Recently, using Schumann resonance intensity variations to monitor changes in both regional and global lightning activity, Williams et al (2021) found a significant rise in lightning activity attributable to thermodynamic disequilibrium in the transition months preceding two “super” El Niño events, namely during 1997-1998 and 2015-2016. Xu et al (2023) suggested that the ENSO phenomenon is the climate driver for the CG lightning activity over land areas of China. Water vapor convergence caused by ENSO phenomenon strengthens the ascending motion, which enhances the CG lightning activity.

Williams et al (2019) assessed the global lightning activity during the two periods of so-called “Big Hiatus” with shorter intervals of temperature being flat or declining with time (1940-1975, 1998-2013). The regional thunder days showed a declining trend in the earlier hiatus (1940–1972), and no trend was found during the more recent hiatus (1998-2014). Chen and Dai (2023) recently examined the occurrence frequency of stable and unstable atmospheric conditions over land in terms of CAPE and convective inhibition (CIN) using homogenized radiosonde data from 1979 to 2020 and found that the atmosphere has become increasingly unstable, that would obviously result in more convective storms, hence more lightning activity. Under the context of global warming, the changes in thermodynamic conditions and microphysical processes both strongly influence lightning activity. The main impact factor should be investigated in different regions, and the combined effect of thermodynamic conditions and microphysical processes should also be considered in the future.

3.2 Trends in thunderstorm day counts

Reliable observations of global lightning have been underway for just over two decades, while thunder day observations at meteorological stations have been available worldwide for more than a century. Thunderstorm day is defined as when one or more thunderstorms or lightning are observed at a station on one day. So thunderstorm day or lightning day has been used for assessing lightning variability on a time scale longer than the typical lifespan of lightning detection networks. Lavigne et al (2019) studied the global thunder days' trend and showed that the majority of the selected regions, such as the Amazon, the Maritime Continent, India, Congo, Central America and Argentina, have exhibited a prominent rise in annual thunder day counts since the 1970s, whereas China, Australia and the Sahel showed a decline in the number of thunder days. They found a positive correlation between the changes of thunder day occurrence and flash density over most regions of the TRMM domain.

Recently, the conspicuous decrease of thunderstorm days in China and the influence of environmental factors have been examined in detail. Zhang et al. (2017) found that the number of days with thunderstorms decreased by about 50% from 1961 to 2010, based on a data set of over 500 observational stations in China. They show a strong correlation of the decrease with the weakening of the east Asian summer monsoon associated with the weakening of dynamic forcing and the decreasing of moisture supply in the warm season. Xue et al (2021) confirmed the significant decreasing of thunderstorm and lightning days based on updated datasets of more than 2,300 (1,737 after screening) stations in China from 1961-2013. They found a strong link of this decreasing trend and the long-term changes of the atmospheric circulation and environmental factors, such as weakening of geopotential height difference between south and north, the westerly jet strength at 200 hPa, zonal wind speed of high level, the value of CAPE, relative humidity at the lower troposphere, and the 0-6 km vertical wind shear.

It is interesting to note that thunderstorm day count over the Tibetan Plateau also declined with an annual average reduction rate of about 1% from 1961 to 2010 (Zou et

al 2018; Tang et al 2021), while lightning activity increased from 1997-2019 (Qie et al 2022). The different variation behavior of lightning and thunderstorm days might be partially a result of the different period considered. However, the thunderstorm day count and lightning flash density are fundamentally different quantities. One thunderstorm may produce a large amount of lightning, although lightning number produced by a thunderstorm is usually not high with a typical flash rate of just 1-4 flashes s^{-1} over the Tibetan Plateau (Qie et al 2005; Zheng and Zhang, 2021).

3.3 Lightning changes in the future

How lightning activity will change in the future with the progression of climate change in the coming decades is the focus of on-going research, due to the serious implications to human societies (Yair, 2018). Based on the meteorological control for thunderstorm electrification and ensuing lightning activity, many attempts have recently been made to establish the relationship between lightning and meteorological proxies in order to predict lightning changes for future emissions pathways.

3.2.1 Lightning proxies based on thermodynamic factors

Using a product of CAPE and precipitation ($CAPE \times P$) as a proxy for lightning, Romps et al (2014, 2019) predicted an increase in future lightning activity in the contiguous United States, stating a 14% increase for any 1K warming. Although this proxy may faithfully replicate the seasonal and diurnal cycle of lightning over the contiguous United States, as well as the distribution of flash rate density over global land areas, it fails to predict the land-ocean lightning contrast (Romps et al 2019). It should be noted that the precipitation at the surface is not a good indicator because electrification is not appreciably effective in the warm rain but occurs through collisions between the ice particles aloft.

Chen et al (2023) suggested an increase in Arctic summer lightning by the end of the century (2081-2100) with the same proxy of $P \times CAPE$ as a lightning proxy in the CMIP5 (Coupled Model Intercomparison Project Phase 5) for a high-emissions scenario (RCP8.5). The lightning increases may lead to the ignition of more forest fires

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and larger burnt areas in the Arctic tundra, expediting the northward migration of boreal trees, which may exacerbate the positive feedback associated with a decrease in albedo and permafrost soil carbon release. Chakraborty et al (2021) used the total column water vapor, CAPE and AOD to characterize lightning over the Indian region and also revealed a uniform increase in lightning occurrence and intensity within two emission scenarios of RCP2.6 and RCP8.5.

3.2.2 Proxy based on cloud microphysics

Based on the relationship between lightning and ice particles in the thundercloud (Takahashi, 1978; Saunders and Peck, 1998), Finney et al (2018) adopted the upward cloud-ice flux (IFLUX) as a proxy of lightning using climate model high-emission scenario (RCP8.5) and projected a significant decrease about -10 % of lightning in the tropics by the end of the century (2100). This prediction is model based with the prediction of a decrease in ice mass flux reaching the upper portions of the troposphere where electrification is known to occur. However, a recent study on extremes in tropical rainfall in a warmer climate suggests the enhancements exceeding the simple predictions based on the Clausius–Clapeyron (Guerreiro et al 2018) and seems to contradict the model predictions of Finney et al (2018). A deep understanding of how thermodynamics and microphysics affect lightning activity is still necessary.

4. Lightning response to changes in aerosols

Although thermodynamic factors are essential for the initiation of deep moist convection, aerosol also play a profound role in thunderstorm intensification and lightning activity. In a thunderstorm, charge is generated by non-inductive electrification process through mutual collisions between graupel particles and ice crystals in the presence of supercooled liquid water leading to their opposite charging (Takahashi, 1978; Saunders & Peck, 1998). The macroscopic separation of positive and negative charge regions in the upwelling airflow generate the electric fields needed to initiate lightning. Increased aerosols provide larger CCN concentration which nucleate

numerous and smaller cloud droplets that are slower to coalesce into raindrops. Cloud water that has not been transformed into raindrops is lifted to above the 0 ° C level to become supercooled water or ice hydrometeors. The phase change from liquid to solid releases latent heat of freezing that invigorates convection and intensifies the electrification (Rosenfeld and Lensky, 1998; Rosenfeld et al 2008, 2014; Mansell and Ziegler, 2013).

4.1 "Boomerang" shape dependence of flash density on aerosols

Activated aerosols invigorate convection and lightning by participating in cloud microphysics under certain aerosol loading level (e.g., Wang et al 2011; Yuan et al 2011; Tan et al 2016), but the trend may flattens or reverses direction when aerosol loading is high.

Based on WLLN lightning data and Aerosol Optical Depth (AOD) from MODIS, Altaratz et al (2010) found that increased concentrations of Amazonian smoke particle resulted in deeper clouds with more frequent lightning, up to mid-level aerosol loading, above which the trend reversed and added aerosol concentration actually decreased lightning activity, creating a "Boomerang" shape dependence of flash density on the AOD.

Proestakis et al (2016) found a similar behavior between lightning activity and AOD anomalies in the Mediterranean, affected by Saharan dust and particles from anthropogenic sources, and the turning point of daily mean AOD is 0.4. Sun et al (2020) suggested a critical turning point of response of lightning activity to the hourly PM_{2.5} prior to the occurrence of thunderstorms in the Beijing metropolitan region. Dayeh et al (2021) found that lightning is positively correlated with the AOD, up to about 0.8, and the trend flattens or reverses direction when AOD is larger than this value. Kucienska et al (2013) also observed the greatest lightning density for moderate values of the aerosol optical depth between 0.2 and 0.35 in maritime clouds near Mexico .

Above "Boomerang" shape dependence of flash density on aerosols may related to both the microphysical effects and radiative effect of aerosols. Aerosols absorb and

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scatter solar radiation and reduce the amount of solar radiation reaching the surface. When the aerosol level is high, its radiative effect may dominate, and thus inhibit convection and as a consequence decrease lightning activity (Li et al 2007; Guo et al 2016). The complicated microphysical and radiative effects of aerosols together with environmental dynamic and thermodynamic processes greatly influence the convection development, electrification and lightning.

4.2 Effects of shipping on oceanic lightning

Thornton et al (2017), based on 12 years of WWLLN data, found an annual increase by approximately a factor of 2 in the frequency of lightning over two major shipping lanes in the Northeast Indian Ocean and South China Sea, caused by the aerosol particles emitted by maritime vessel traffic enhance lightning through microphysical effects. These findings are confirmed by Liu et al (2020) with data from two additional lightning detection networks: Earth Networks Total Lightning Network (ENTLN) and Global Lightning Detection Network (GLD360). The 2 years (2017–2018) data shows that the mean stroke density, intracloud/CG flashes (IC/CG) ratio, positive IC (+IC) fraction, and positive CG (+CG) fraction in polluted conditions are all larger than in unpolluted counterparts. However, Peterson (2023) suggested that these increased lightning over certain shipping lanes is more likely a direct effect of the ship as a grounding metal object protruding from the sea surface into the high electric field of the thundercloud, thus enhancing the likelihood of a lightning discharge. This result highlighted the central role of direct ship interactions and suggested an alternative interpretation for the observed lightning enhancements over shipping lanes.

4.3 Aerosol effects under different thermodynamic conditions

Although it is challenging to distinguish the aerosol influence from the thermodynamic/dynamic effects, previous research has tried to deal with this difficult issue. Wang et al (2018) investigated the relative roles of aerosols and dynamics-thermodynamics on lightning using 11-year TRMM-LIS lightning and confirmed that

the lightning rate varies with AOD in a boomerang shape (Altaratz et al 2010): first increasing with AOD, tailing off around AOD=0.3, and then decreasing for dust and flattening for smoke aerosols (as shown in Fig. 2), reflecting the corresponding different thermodynamics (in particular humidity and CAPE) and aerosol types. The aerosol invigoration effect could be sustained in the entire range of AOD by high humidity and CAPE.

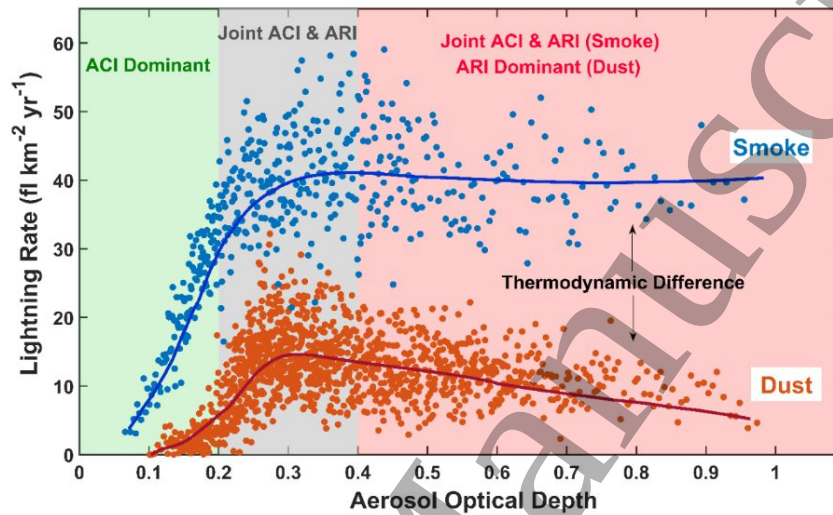


Fig. 3 Lightning flash rate as a function of AOD in the dust- (orange points) and smoke-dominant regions (blue points). Turning points in the boomerang shapes are around AOD=0.3. Aerosol–cloud interactions (ACI) play a dominant role in lightning activity under relatively clean conditions (green zone). As AOD exceeds 0.3, both ACI and aerosol–radiation interaction (ARI) effects come into play with different magnitudes. (Adopted from Wang et al 2018).

Zhao et al (2020) investigated the joint effects of aerosol, thermodynamic, and cloud-related factors on CG lightning in the Tibetan plateau region, where the CCN concentration is usually low. They suggested that the higher aerosol loading increases total column liquid water and total column ice water, and thus enhancing charging efficiency and CG lightning frequency. Wang et al (2022) further showed the different aerosol effects on CG lightning during daytime (inhibiting lightning) and nighttime (no obvious effect) over the polluted Sichuan Basin in the east of the Tibetan Plateau and explained this by aerosol-radiation interactions.

Altaratz et al (2017) examined the association between aerosol loading and lightning on a global scale based on MODIS-derived AOD and WLLN data and concluded that lightning activity was always significantly higher on polluted days than

on clean ones for all CAPE and vertical velocity at the 400 mb pressure level, suggesting that the higher lightning amount detected in polluted conditions could be a direct consequence of the aerosol loading effect on cloud microphysical processes for specified thermodynamic conditions. Stolz et al (2015, 2017) found that the total lightning density within individual convective features from TRMM was enhanced by up to 600% and the average height of 30 dBZ echoes was enhanced by up to 2–3 km with increasing normalized CAPE and CCN concentration ($D \geq 40 \text{ nm}$) for a fixed warm cloud depth. The relationship between lightning and radar reflectivity varies as a function of the CCN concentration ($D \geq 40 \text{ nm}$) for a fixed thermodynamic environment after a statistical regression analysis. Hu et al (2019) quantitatively assessed the impact of CCN on lightning via tracking convective cells based on data from polarimetric radar and the Lightning Mapping Array (LMA), and they found that increasing the CCN concentration from ~ 400 to an optimum of $\sim 1,000 \text{ cm}^3$ enhanced lightning activity by an order of magnitude. A further addition of CCN particles actually decreased flash rates, as observed by others. Augmented CAPE increased lightning only under low CCN (e.g., less than 500 cm^3).

Pinto et al (2020) showed a non-linear correlation between lightning and aerosol concentrations during the COVID-19 shut-down period, which was characterized by a significant decrease in air pollution. Based on ENTLN data, Pérez-Invernón et al (2021) compared the pollutant concentration and lightning activity before and after the COVID-19 lockdown in northern Italy and found that 60% of the decrease in lightning occurrence could be attributed to meteorological conditions, while 40% was attributed to the reduced aerosol emissions.

4.4 Aerosol composition effects

The aerosols type or atmosphere component can also impact lightning activity. Different types of aerosols have different physical and chemical properties, which may change the thermodynamic and water vapor environment in their vicinity during the formation and development of water drops and ice crystals, affecting the electrification

process. Zhao et al (2020) suggested that the relative concentration of sulfate aerosols was more strongly associated with lightning than the total aerosol concentration. Ren et al (2019) did not find a significant linear relationship between aerosols and lightning occurrence rate, but a positive correlation between absorbing aerosols and lightning rate. Because the absorbing aerosols at the top of the boundary layer formed a warm layer at high altitude, local convection was inhibited, led to the accumulation of CAPE, and promoted a more intense convection to develop.

Hu et al (2019) suggested that urban ultrafine aerosols strengthen to the storm electrification. Pan et al (2022) confirmed that fine and coarse soluble aerosols show opposite impacts on the convective invigoration and electrification of thunderclouds. Fine aerosols suppress coalescence and invigorate convection and cloud electrification, while coarse sea spray aerosols weaken convection and inhibit electrification, and consequently suppress lightning activity in oceanic areas. Sun et al. (2023) suggested that the dust aerosols formed more ice-phase particles and supercooled water in the mixed-phase area of the thunderstorm clouds, resulting in a high proportion of +CG lightning.

Lang et al (2014) found that Pyro-cumulonimbus clouds (Pyro-Cb), developed over wildfires, produce lightning whenever the smoke plumes reach up above 10 km MSL (approximately -45°C), due to the transition from pure smoke/ash to frozen hydrometeors based on Doppler and polarimetric radar observations (LaRoche and Lang, 2017). Liu et al (2021) showed that under conditions where thermodynamics does not support frequent lightning activity over the ocean, the number of lightning strikes increased by 270% during the wildfire season, which is higher than the 73% increase over land. The increase in aerosols leads to higher concentrations, smaller sizes, and stronger mixed phases of ice crystals, mainly enhancing the +IC and negative CG of ocean convection.

The satellite- and ground-based lightning sensors detected astonishing flash rate of 2,600 flash min^{-1} during the most violent eruption of the underwater Hunga Tonga-Hunga Ha'apai volcano in January 2022 (Van Eaton et al 2023). Bor et al (2023) concluded that lightning during the volcano eruption dominated the naturally occurring

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global lightning activities for a period of at least 1 hour based on Schumann resonance measurements from six stations on four continents. The exceptional emplacement of water substance into the atmosphere by this eruption might be responsible for the exceptional lightning activity of this Tonga eruption. However, this event represents an extreme case and is not representative of the wider aerosol effects on lightning activity on regional and global scales.

4.5 Numerical simulations of aerosol effects

In recent years, many numerical simulations have studied the role of aerosol in the thunderstorm charge structure and lightning activity. The simulations support the influence of aerosol on the lightning flash rate, even on the lightning type and polarity, through impacting the electrification of hydrometeors and thunderstorm charge structure, which are closely related to the microphysical processes in the mixed phase region (e.g., Mansell & Ziegler, 2013; Zhao et al 2015; Tan et al 2017; Lynn et al 2020; Sun et al 2021, 2023, Hazra et al 2022).

Combined the spectral bin microphysics scheme in WRF with the prognostic GoCart model in WRF, Lynn et al (2020) found that the urban aerosols increased lightning rates along the Israeli coastline, while the combination of dust and urban aerosols drives the most intense lightning activity. Different nucleation schemes in simulations affect the participation of aerosols in the electrification process. Sun et al (2024) studied the effects of aerosols on the electrification and lightning based on the WRF-ELEC model (Fierro et al 2013) coupled with a sophisticated explicit charging parameterization (Saunders & Peck, 1998) and three-dimensional lightning scheme (Mansell et al 2005; MacGorman et al 2001). They suggested that both IC, CG flashes and fraction of +CG are all increased by adding CCN concentrations through modifying the microphysical processes. Increased aerosols enhance the number of cloud droplets and ice-phase hydrometeors, providing sufficient charge separation and leading to an inverted charge structure with a middle main positive charge region forming from positively charged graupel in the presence of higher supercooled cloud water content.

Inverted polarity of charge structure is conducive to positive lightning flashes. In the scenario of low aerosol concentration, the positive charge density in the upper layer is much lower resulting from smaller ice-particle content, leading to barely any +CG strokes.

5. Summary

Available observational data has indicated significant lightning activity increase in some regions, such as the Arctic, south Asia, and the Tibetan Plateau, in the context of global warming. Most of the model results also projected an increase in future lightning, such as the Continental US, the Arctic, and south Asia. From a thermodynamic perspective, a warmer atmosphere and higher land/sea surface temperatures will lead to increased CAPE values and enhanced air instability, a trend which has been proven recently (Chen and Dai, 2023), and thus is expected to promote stronger and deeper convection and increase cloud electrification and lightning activity.

The relationship between lightning and aerosols generally shows a "boomerang" change pattern. Activated aerosols invigorate convection and lightning by participating in cloud microphysics under certain aerosol loading levels. However, this relationship depends on the environmental thermodynamic conditions. The atmosphere components also affect the cloud microphysical process and lightning.

Although substantial progress has been made in recent years in understanding relationships between lightning activity and climatic changes in temperature and aerosol conditions, a number of questions still remain open. The first and perhaps most interesting question concerns the way lightning patterns and characteristics will change in the future. The changes in aerosol concentration caused by human activities will affect changes in temperature and thermodynamic environment, these factors together will lead to uncertainties in future lightning patterns or its changing rate. The stricter regulation on anthropogenic emissions (for example, the transfer to low-sulfur fuels in shipping, known as IMO20 which already showed considerable effects on marine cloudiness, and may actually reduce the enhancement of lightning observed over shipping lanes; Yuan et al., 2023) and the transition to renewable energy

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production will eventually decrease the amounts of aerosol particles in certain regions with significant cloud microphysical consequences, as already observed above the oceans (Diamond et al 2023). This will affect lightning occurrence above the oceans and over major metropolitan areas everywhere. Based on deep understandings of the relationship between lightning and environmental thermodynamic conditions, and their descriptions in advanced numerical models will greatly improve the success of model projections in lightning response to temperature.

The second outstanding question concerns the intrinsic mechanisms in the prominent land/ocean lightning contrast. The combined effect of both thermodynamics and aerosol concentrations and properties may play a significant role in the land/ocean lightning contrast because of their impact on updrafts and microphysical processes of the thunderstorm, but their respective quantitative contributions still need to be addressed. Other factors, the terrain topography, the cloud base height, as well as the different shapes of updraft buoyancy profiles over land compared to the ocean, may also contribute to the land-ocean contrast in lightning and thunderstorm. The different behaviors and effects of land/ocean lightning and thunderstorms are also subjects of interest. Lu et al. (2022) found that negative cloud-to-ground (CG) strokes produced by oceanic thunderstorms are more readily to produce sprites. Another interesting aspect of this land-ocean contrast is the fact that very bright and powerful flashes, known as superbolts (Holzworth et al., 2019; Peterson, 2023), occur more frequently over oceanic and coastal regions as compared with land areas. This reflects differences in thunderstorm dynamics, where oceanic convection is less vigorous than continental, yet still manages to produce vast amounts of electric charge that accumulates for long periods of time and is released as extremely energetic flashes.

The third question concerns the robustness of response of lightning to temperature and aerosols over a longer time scale. The response of lightning to temperature and aerosols over a longer time scale, as well as the robustness of observed and projected lightning trends, require continuous monitoring with consistent spatial and temporal detection efficiency over a longer time scale. The global lightning observations in geostationary orbit, Geostationary Lightning Mappers (GLMs, Goodman et al, 2023;

Mach, 2020; Peterson and Lay, 2020) on GOES-16 and 17, Lightning Mapping Imager (LMI; Cao et al., 2021; Chen et al 2021; Zhang et al 2023) on Fengyun-4, and other potential satellites, as well as the continuously advancing ground-based global lightning detection network, are providing or will provide valuable global lightning data to hopefully address these issues.

In conclusion, figuring out the various facets of the mysteries of lightning response to temperature and aerosols in the context of global climate change is obviously a primary subject of interest, and needs to be continuously studied.

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Data availability statement

No new data were created or analysed in this study.

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